Xi Li

Machine Learning Researcher — Biomedical AI and Computational Pathology

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Seeking Full-Time Research and Engineering Roles

Profile

Ph.D. researcher in Computer Science at UC Irvine, developing multimodal AI that connects tissue morphology, spatial gene expression, and disease biology. First author of **STMoE** and **sMMC-22M**, with experience in organ-specialized prediction, cell-aligned data at tens-of-millions scale, and interpretable drug-target reasoning through **TargetSage**. Research spans model design, data alignment, and cross-sample evaluation, complemented by an internally deployed generative-AI platform at ByteDance and end-to-end mentorship of student-led research.

Education

University of California, Irvine	*Ph.D., Computer Science*	Sep 2023 – Present
Tsinghua University	*M.S., Computer Science and Technology*	Sep 2019 – Jun 2022
University of Science and Technology of China	*B.S., Information Security*	Sep 2015 – Jun 2019

Expected Ph.D. graduation: June 2027.

Research Experience

University of California, Irvine — **Irvine, CA**
Ph.D. Student Researcher — *Sep 2023 – Present*

- **STMoE — histology-to-expression modeling.** Led development of a multi-scale mixture-of-experts model for **single-cell gene-expression prediction across 25 organs**. Combined **single-cell and tissue-context crops** with frozen UNI2-h representations, soft contrastive alignment, and organ-routed retrieval; restricted the molecular reference bank to training data.
- **Cross-sample generalization.** Retained **70% of predictive performance** when generalizing to unseen samples.
- **Data engineering and leakage control.** Built cell-aligned data and benchmarks for **CellST-24M** and **sMMC-22M**; tracked study/slide provenance, assay, and coordinate metadata, distinguishing native Xenium cells from derived Visium HD targets. The **expanded sMMC collection** contains **28.3M cell-aligned targets across 25 tissue categories**; the formal paper retains the frozen-release name sMMC-22M.
- **Evaluation design.** Constructed in-domain, cross-patient, and cross-platform benchmarks with training-only reference banks to assess transfer while limiting molecular-reference leakage.
- **TargetSage — interpretable drug-target discovery (co-author).** Responsible for **disease-state verification and multi-target investigation** within an evidence-grounded LLM-reasoning framework. The overall model improved **macro-average Adjusted F1 by 37% relative to the strongest of 10 baselines**, across 15 tasks and 19,032 genes.
- **Extension to spatial perturbation modeling.** Extended the TargetSage reasoning approach to spatial perturbation research, connecting disease-state context and multi-target hypotheses with perturbation-conditioned tissue responses.
- **Spine — spatial response modeling (first author).** Developed a formulation that separates tissue context from perturbation-conditioned response fields in spatial genomics; manuscript under review.

Multimedia Lab, Tsinghua University — **Beijing, China**
Researcher, AI Broadcast Production — *Oct 2021 – Sep 2022*

- Built an audio–visual cross-attention system for automated camera switching, crop selection, and virtual camera motion; **deployed at China's National Centre for the Performing Arts** for live orchestral production.

Technical Skills

Modeling Multimodal learning, transformers, mixture-of-experts, contrastive learning, LLM reasoning, positive–unlabeled models

Engineering Python, PyTorch, multi-GPU experimentation, image–omics alignment, dataset QC, grouped evaluation and reproducible benchmarking

Mentoring and Open-Source Leadership

Google **Summer of Code (GSoC)** — **2025 & 2026**
Project Mentor

- Provided **end-to-end mentorship to two students** across the 2025 and 2026 projects, guiding problem formulation, implementation, experiments, and paper development; both projects have code repositories.
- Guided a student to publish **one student-first-author paper at BIBM**.
- Projects: **2026: Omni-ST** — instruction-driven multimodal spatial modeling; **2025: Disentangled Generation and Editing of Pathology Images** — controllable pathology-image generation and editing.

Industry and Product Experience

Hantao Zhiying **China**
Chief Product Lead *2022 – 2023*
- Owned product strategy and delivery for AI imaging products, translating product goals into feature definitions, user-experience requirements, and implementation priorities.
- Coordinated research, engineering, and design teams across technical planning and deployment.

ByteDance AI Lab **Beijing, China**
AI Product Manager Intern *May 2021 – Dec 2021*
- Incubated **Alchemy**, an internal image-generation platform; drove product scoping and internal launch for **Fanqie Novel book-cover creation**. Coordinated research, engineering, and design to bring image-generation and editing models into the product workflow.
- Participated in the **Chimera** digital-human project and owned automotive-client engagement, presenting **digital-human in-car assistant** solutions and supporting solution sales.

ByteDance — Feishu/Lark **Beijing, China**
Data Product Intern *Jun 2020 – Nov 2020*
- Used behavior analytics and relevance analysis to examine enterprise document and knowledge-retrieval workflows; translated findings into data-informed search-product design for Feishu/Lark.

Nolibox / Computational Aesthetics **Beijing, China**
Partner and Co-founder *Mar 2020 – Aug 2020*
- Developed an automatic logo-generation system for AI-assisted visual design.
- Built font-stroke generation tools to support typeface-design workflows.

Company milestone (post-tenure): Series A financing of tens of millions of RMB (2024).

Selected Publications and Manuscripts

Publication links: xenali96.github.io/publications.html.

Published / accepted

1. **Xi Li**, Yaqi Hu, Ziheng Duan, Chaoyang Wang, and Jing Zhang. "STMoE: Multi-Scale Mixture-of-Experts for Single-Cell Gene Expression Prediction from Histology." **CIKM 2026**, accepted. *First author.*

2. **Xi Li**, Yaqi Hu, Ziheng Duan, Diptanshu Sikdar, Yang Liu, and Jing Zhang. "sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics." **NeurIPS 2026**, accepted poster. *First author.*

3. Ziheng Duan*, **Xi Li***, Zhiqing Xiao, Rex Ying, and Jing Zhang. "MUSE: A Multi-Slice Joint Analysis Method for Spatial Transcriptomics Experiments." **CIKM 2025**. **Co-first author.**

4. Ziheng Duan, Tong Wu, **Xi Li**, Simon Dongbo Sun, Chaoyang Wang, Junhao Liu, Martin Renqiang Min, and Jing Zhang. "TargetSage: Identifying Therapeutic Target Genes with Interpretable and Robust LLM Reasoning." **NeurIPS 2026**, accepted poster.

5. **Xi Li**, Ziheng Duan, Yi Dai, Siwei Xu, and Jing Zhang. "iHAST: Integrating Hybrid Attention for Super-Resolution in Spatial Transcriptomics." **BMVC 2024**. *First author.*

6. Ziheng Duan, **Xi Li**, Zhuoyang Zhang, James Song, and Jing Zhang. "DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion." **ICIP 2025**.

Manuscripts

7. Liyun Lin, Shuozhen Bao, **Xi Li**, Isaiah Yim, Dejiang Wang, Gaocai Li, Xiangjun Di, Shujie He, Junyi Xu, Kathryn Miller-Jensen, Jing Zhang, and Rong Fan. "Spatially Indexed Single-Cell Morphology and Transcriptome Profiling by DBiC-seq." **Nature Communications**, under review.

8. **Xi Li**, Ziheng Duan, Yi Dai, Yang Liu, and Jing Zhang. "Context Is Not a Side Note: A Support-Calibrated Reliability Framework for Biomedical Foundation Models." **KDD 2027**, under review. *First author.*

9. **Xi Li**, Ziheng Duan, Simon Dongbo Sun, Martin Renqiang Min, Yang Liu, and Jing Zhang. "Spine: Disentangling Tissue Context and Perturbation-Conditioned Response Fields in Spatial Genomics." **KDD 2027**, under review. *First author.*

10. "CCR2 Deficiency Attenuates Flow-Induced Carotid Artery Remodeling." **Vascular Pharmacology**, under review. Co-authors include Fu Gao and **Xi Li**; corresponding author: Yang Liu.

11. Fu Gao, Ian Mason, Mingze Dong, Yao Lu, Di Zhang, Xing Lou, **Xi Li**, Irbaz Hameed, Mingyu Yang, Mei Zhong, Markus Krane, Giovanni Ferrari, George Tellides, Yang Liu, Rong Fan, and Arnar Geirsson. "Cross-Species Multi-Omics Identifies a Conserved Fibrogenic Valvular Interstitial Cell Population across Mitral Valve Diseases." **Circulation**, submitted.